

A PUZZLE ABOUT A PROMISE

A friend offers to sell you a paper. It promises Tk. 100 each year for three years and Tk. 1,000 at the end. How much should you pay? Not Tk. 1,300, the total promised, since later money is worth less today. The right price is the present value of everything the paper promises. This chapter applies Chapter II to real securities.

THE IDEA OF VALUE

A **security** means a paper or record that gives its holder a claim on money. **Intrinsic value** means what a security is worth, found from the present value of the cash it is expected to pay, discounted at the return investors require. If the market price is below this value, the security looks cheap; if above, it looks costly.

BONDS

A **bond** means a long-term loan to a firm or government, divided into pieces that can be traded. Its main features are these. The **face value** is the sum repaid at the end, often Tk. 1,000. The **coupon rate** is the yearly interest as a percentage of face value. The **maturity** is the date of repayment. Other features include the claim on assets if the issuer fails, and whether it can be redeemed early.

The price of a bond follows the rule of Chapter II. Price = present value of the interest payments (an annuity) + present value of the face value (a single sum), both at the required return. The **required rate of return** is set by the market for bonds of that risk.

WORKED EXAMPLE: VALUE OF A THREE-YEAR BOND

GIVEN. face value Tk. 1,000; coupon rate 10% paid yearly; 3 years; required return 12%.

FIND. the bond's value.

FORMULA. Value = interest x annuity factor + face value x single-sum factor.

STEPS. Interest = $1,000 \times 10\% = \text{Tk. } 100$. Annuity factor at 12% for 3 years = 2.40183. Single-sum factor = 0.71178. Value = $100 \times 2.40183 + 1,000 \times 0.71178 = 240.18 + 711.78$.

ANSWER. The value is Tk. 951.96.

CHECK. The coupon (10%) is below the required return (12%), so the bond must be worth less than its face value of 1,000. It is.

BOND STATUS AND YIELD

The **bond status** tells whether the bond sells at a discount (below face value), at par (equal) or at a premium (above). Compare the coupon rate with the required return. If the coupon is lower, the bond sells at a discount, because buyers must be compensated. If equal, at par. If higher, at a premium. When market rates rise, bond prices fall, and when they fall, prices rise.

The **yield to maturity** (YTM) is the yearly return earned by buying at the market price and holding to maturity. It is the discount rate that makes the present value of all payments equal to the price. In the example, a price of Tk. 951.96 gives a YTM of 12%. At a required return of 8% the same bond is worth Tk. 1051.54, a premium. At 10% it is worth Tk. 1,000, which is par.

A paper that promises money has a price.

Value is the present value of expected cash at the required return.

Security. A paper or record that gives its holder a claim on money from a borrower or a company.

Intrinsic value. What a security is worth, found from the present value of the cash it is expected to pay.

A bond is a loan split into small, tradable pieces.

Bond. A long-term loan to a firm or government, with fixed payments, that can be bought and sold.

Face value. The sum printed on the bond and repaid at maturity; also called par value.

Coupon rate. The yearly interest rate on the face value, set when the bond is issued.

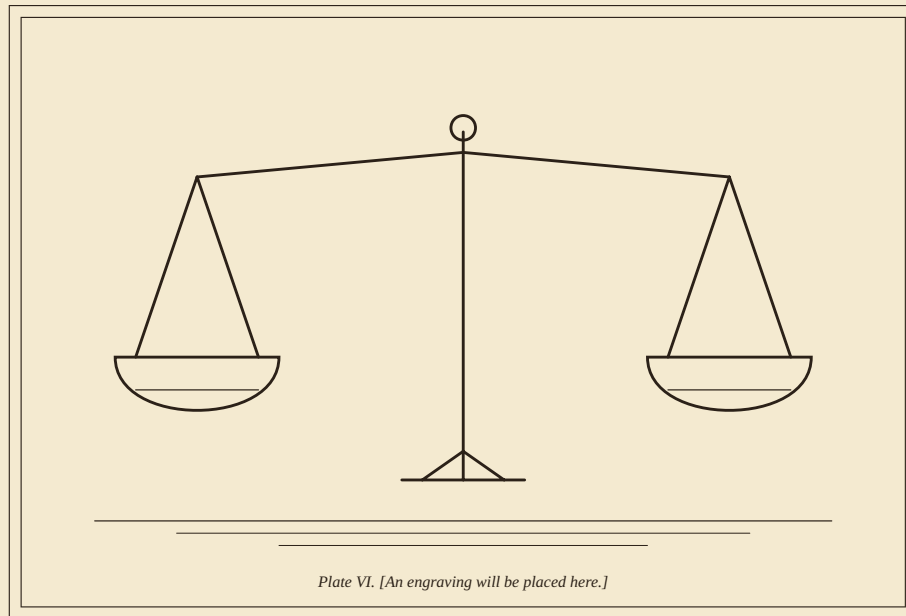
Maturity. The date on which the bond is repaid.

Required rate of return. The yield investors need from a bond of this risk, set by the market.

The bond sells at a discount, par or premium.

Bond status. Whether a bond sells below, at or above its face value.

Yield to maturity. The yearly return an investor earns by buying a bond at its price and holding it to maturity.



CHAPTER VI

Valuing Bonds and Shares

What is a promise of future money worth today?



This chapter explains how securities are valued, how a bond is priced from its features, what its status and yield mean, how a bond differs from a share, and how a share is valued.

can raise the return on the owner's money when business is good. From the lender's side, high debt means a higher chance of not being repaid. From the manager's side, the task is to choose a level that gains from leverage without inviting failure.

Exceptions. DOL is different at each sales level, and it cannot be found at break-even, where EBIT is zero. Leverage figures assume that price, variable cost and fixed cost stay unchanged.

CLOSING THE CHAPTER

SUMMARY

Fixed costs act as a lever. Operating leverage comes from fixed operating costs; financial leverage from fixed financing costs; total leverage combines them as $DTL = DOL \times DFL$. Leverage magnifies gains and losses and therefore adds to risk.

THINK IT THROUGH

A firm sells 2,000 units at Tk. 60, with variable cost Tk. 40 a unit, fixed operating cost Tk. 20,000 and interest Tk. 5,000. Find DOL, DFL and DTL.

WHERE PEOPLE GO WRONG

Students use sales instead of contribution in DOL. Others forget that DTL is the product, not the sum, of DOL and DFL.

IN HISTORY

The Greek writer Pappus, who lived in the fourth century, reported that Archimedes had claimed he could move the Earth if given a place to stand. Whatever the truth of the saying, it made the lever a lasting picture of small force producing large effect, and financiers later borrowed the word.

REVIEW QUESTIONS

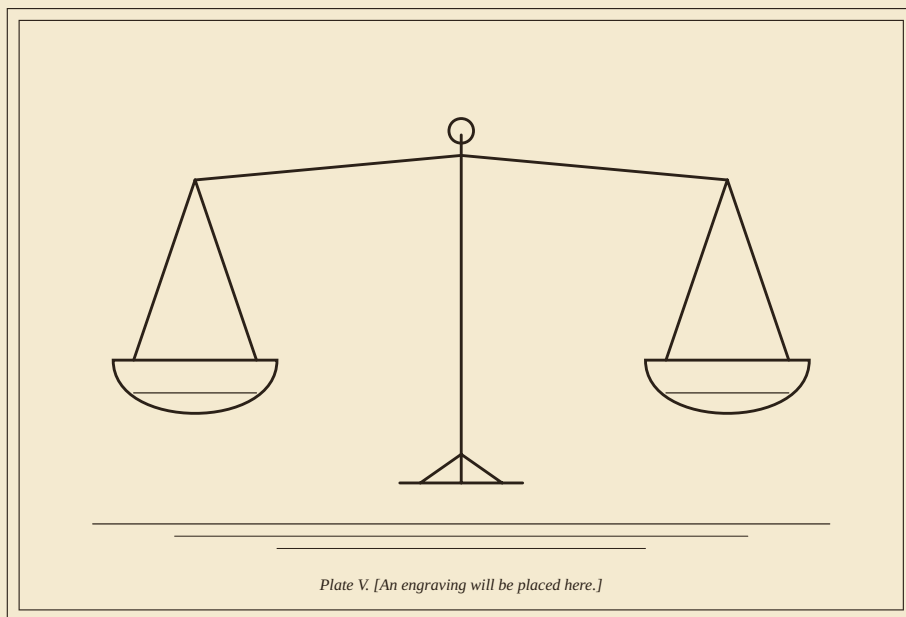
1. Define operating, financial and total leverage.
2. Why is leverage significant for a firm?
3. Solve the Think it through problem.

IN HISTORY

In Holland in 1636 and 1637, contemporaries reported that prices for tulip bulbs rose sharply and then collapsed. Historians still debate how large and how harmful the episode was, but it remains a warning about prices that float far above any present value of expected cash.

REVIEW QUESTIONS

1. What is a bond? Name its key features.
2. How does the required return decide bond status?
3. Solve the Think it through problem.
4. How does a bond differ from a share?



CHAPTER V

Leverage

A small push on one end lifts a large weight on the other.



This chapter explains operating, financial and total leverage, shows how to measure each one, and discusses why leverage can magnify both profit and trouble.

Orders do not pay wages; cash does.

Working capital is the money tied up in daily operations.

Working capital. The money a firm needs for daily operations, held in cash, stock and money owed by customers.

Liquidity. How quickly and easily an asset can be turned into cash without loss.

Financing is named by how long the money is borrowed.

Trade credit. Goods bought now and paid for later, which is a free or cheap loan from the supplier.

Match the life of the money to the life of the asset.

Matching principle. Financing an asset for about as long as the asset will be used.

A discount for early payment has a hidden interest rate.

Cash discount. A reduction in price offered for payment within a short time.

A PUZZLE ABOUT A FULL ORDER BOOK

The Bakery receives a large order from a hotel. To bake it, the Bakery must buy flour on Monday and pay wages on Friday. The hotel will pay in forty days. For forty days the Bakery spends money and receives none. If it cannot find that money, the best order of the year turns into a crisis. Short-term money is about surviving this gap.

WORKING CAPITAL

Working capital means the money a firm needs for daily operations. It is held in current assets: cash, stock of goods and the money customers owe. **Liquidity** means how quickly an asset can be turned into cash. Too little working capital risks failure to pay bills. Too much leaves money idle that could earn a return elsewhere.

SHORT, MEDIUM AND LONG FINANCING

Financing is grouped by maturity. Short-term financing is repaid within one year. Medium-term, or intermediate, financing runs from one year to about five years. Long-term financing runs for more than five years. The limits vary between books and countries, so the grouping is a guide. The next chapter studies long-term sources.

Common sources of short-term finance are these. **Trade credit** means goods bought now and paid for later. Bank overdrafts and short bank loans give quick money at a stated interest. Accrued expenses, such as wages owed but not yet paid, are a form of free credit. Firms may also borrow against money their customers owe.

THE MATCHING PRINCIPLE

The **matching principle** says to finance an asset for about as long as it will be used. A delivery van used for five years should be paid for with money repaid over about five years. Seasonal stock should be financed with short loans. If a firm finances a long-lived machine with an overdraft, the lender may demand repayment before the machine has earned the money. If it finances stock with long-term debt, it pays interest on money it does not need all year.

THE COST OF TRADE CREDIT

A supplier may say "2/10 net 30". It means: take 2% off if you pay within 10 days; otherwise pay the full sum by day 30. The **cash discount** is a reward for paying early. Not taking it is like borrowing from the supplier at a cost. Cost = [discount / (100 - discount)] x [360 / (final day - discount day)].

WORKED EXAMPLE: THE COST OF GIVING UP A DISCOUNT

GIVEN. terms 2/10 net 30. A bank loan costs 14% a year.

FIND. the yearly cost of trade credit, and whether to take the discount.

FORMULA. Cost = [d / (100 - d)] x [360 / (N - D)], where d is the discount per cent, D the discount day and N the final day.

STEPS. d / (100 - d) = 2 / 98 = 0.0204. N - D = 30 - 10 = 20 days, so 360 / 20 = 18. Cost = 0.0204 x 18 = 36.73%.

CHECK. Y's returns lie twice as far from the average as X's, and its standard deviation is twice as large.

You should now be able to say: *standard deviation tells me how widely returns spread; a larger value means more risk.*

Exceptions. Standard deviation compares well only when expected values are equal; when they differ, use the coefficient of variation. Standard deviation treats gains and losses alike. Past results do not guarantee future ones.

RISK AND THE RETURN INVESTORS DEMAND

Investors dislike risk, so they require extra reward for bearing it. The **risk premium** is that extra return. The **required return** is the minimum return needed to accept an investment: required return = risk-free return + risk premium. The risk-free return is what a safe government security pays.

The Capital Asset Pricing Model (CAPM) sets the premium by the risk that cannot be removed. Required return = risk-free return + beta x (market return - risk-free return). Beta shows how strongly an investment moves with the whole market. With a risk-free return of 5%, a market return of 11% and a beta of 1.5, the required return is $5 + 1.5 \times (11 - 5) = 14\%$. A beta above 1 means more risk than the market as a whole.

MARKETS AND INFORMATION

An **efficient market** is one in which prices already reflect the available information. Three forms are named. In the weak form, prices reflect all past prices, so studying old price charts brings no gain. In the semi-strong form, prices also reflect all public information, such as published accounts. In the strong form, prices reflect all information, even private. Most evidence supports the weak and, to a large degree, the semi-strong form.

CLOSING THE CHAPTER

SUMMARY

Return is the gain on an investment; risk is the chance that the gain differs from what is expected. Business and financial risk are the two main kinds in a firm. Expected value, standard deviation and the coefficient of variation measure risk. Investors demand a premium for risk.

THINK IT THROUGH

Asset M has expected return 10% and standard deviation 4%. Asset N has expected return 20% and standard deviation 6%. Which is less risky for each unit of return?

WHERE PEOPLE GO WRONG

Students forget to take the square root, or compare standard deviations when expected values differ. Others think a high return is a sign of low risk.

IN HISTORY

In 1688, according to the records, Edward Lloyd's coffee house in London was already a meeting place for shipowners, captains and merchants. Insurers gathered there to share the risks of sea voyages, and from this habit grew the market called Lloyd's.

REVIEW QUESTIONS

1. Define short-term, medium-term and long-term financing.
2. Find the cost of trade credit for 2/15 net 45.
3. Solve the Think it through problem.
4. What is float and what are its parts?

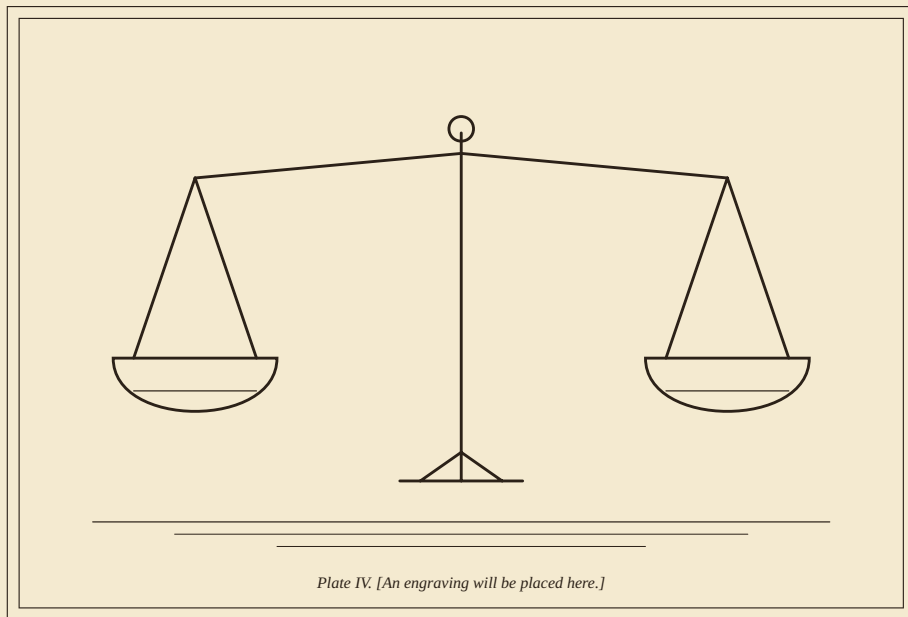
More risk must be paid for with more return.

Risk premium. The extra return investors require for taking risk instead of a safe investment.

Required return. The minimum return an investor needs to accept an investment.

Prices in an efficient market already reflect information.

Efficient market. A market in which prices already reflect the available information.



CHAPTER IV

Risk and Return

Two roads reach the same place; one is safer.



This chapter explains return and risk, the kinds of risk a firm faces, how to measure risk with expected value, standard deviation and the coefficient of variation, and how risk sets the return that investors demand.

New firms have little to offer lenders.

Six features let us compare sources.

Debt is repaid with interest on fixed dates.

Term loan. A loan from a bank or other lender for a fixed period, repaid in instalments.

Debenture. A long-term bond issued by a company, usually paying fixed interest.

Owners accept the highest risk and expect the highest return.

Preferred stock. Shares that pay a fixed dividend and rank before ordinary shares, usually without a vote.

Common stock. The ordinary shares of a company; holders vote and share in the profit left after others are paid.

Retained earnings. Profit that the firm keeps and reinvests instead of paying out as dividends.

A PUZZLE ABOUT STARTING OUT

A new firm has no profits, no record and few assets. Yet it needs a building and machines that will serve for ten years. Lenders hesitate and owners have limited savings. Where does the money come from, and what does each source ask in return? Long-term finance is the answer, and it is especially important for a new business, since a start-up's assets earn slowly and cannot be paid for from short loans.

FEATURES OF ANY LONG-TERM SOURCE

Each source can be compared on six points. Its claim on income: who is paid first. Its claim on assets if the firm fails. Its control: whether the supplier gets a vote. Its cost: the return the supplier requires. Its maturity: when repayment is due. Its risk to the firm: what happens if payment is missed. Debt is cheaper and has no vote but must be paid. Ownership money costs more but has no fixed payment.

BORROWED MONEY

A **term loan** means a loan for a fixed period, usually from one to ten years, repaid in instalments. It is common for buying machines. The lender often asks for security, which is an asset that the lender may take if the loan is not repaid. A **debenture** is a long-term bond issued by a company, which is how a company borrows from many lenders instead of one. The methods of debt thus include bank term loans, bonds and debentures.

Interest on debt is usually a cost that reduces taxable profit, so the real cost of debt is lower than the interest rate. After-tax cost of debt = interest rate $\times$ (1 - tax rate). If a firm pays 12% and the tax rate is 30% (an assumption for illustration; check the current rate), the after-tax cost is $12 \times 0.70 = 8.4\%$.

OWNERSHIP MONEY

Preferred stock means shares that pay a fixed dividend and are paid before ordinary shareholders. They are a mixture: like debt in their fixed payment, and like ownership in that a missed dividend does not bankrupt the firm.

Common stock, also called ordinary shares, means the shares of the true owners. Holders vote, receive what is left after others are paid, and bear the most risk. Their required return is therefore the highest.

Retained earnings means profit that the firm keeps instead of paying as dividends. It is the cheapest source to obtain, with no fees and no new owners, but it is not free: the owners could have used the money elsewhere, so they expect it to earn at least their required return.

THE SOURCES COMPARED

Source	Paid	Vote	Cost to firm	If payment missed
Debt	Fixed interest, first	No	Lowest after tax	Lender may take assets
Preferred stock	Fixed dividend, second	Usually no	Middle	Dividend is owed, firm survives
Common stock	What remains, last	Yes	Highest	No legal default
	Not applicable			Not applicable

WORKED EXAMPLE: MACHINES A AND B AT A DISCOUNT RATE OF 10%

GIVEN. cost Tk. 10,000 each. A: 5,000, 5,000, 5,000. B: 2,000, 4,000, 10,000. Rate 10%.

FIND. payback period and NPV of each; which to prefer.

FORMULA. Payback = full years + (cost still unpaid / cash flow of the next year).
NPV = sum of PV of inflows - initial cost.

STEPS. A pays back in $10,000 / 5,000 = 2.0$ years. B returns 2,000 and 6,000 by the end of years 1 and 2, so 4,000 is unpaid; $4,000 / 10,000 = 0.4$, so payback is 2.4 years. NPV of A = $5,000 \times 2.48685 - 10,000 = 12,434.26 - 10,000 = \text{Tk. } 2,434.26$. NPV of B = $2,000 \times 0.9091 + 4,000 \times 0.8264 + 10,000 \times 0.7513 - 10,000 = 12,637.11 - 10,000 = \text{Tk. } 2,637.11$.

ANSWER. A has the shorter payback (2.0 against 2.4 years), but B has the higher NPV. The two methods disagree. NPV counts timing and all cash flows, so B is preferred as the better addition to wealth.

CHECK. B's large final payment of 10,000 is worth Tk. 7,513.15 today, which more than makes up for its slower start.

WORKED EXAMPLE: THE BAKERY'S SECOND OVEN AT 12%

GIVEN. cost Tk. 5,00,000. Inflows: Year 1 1,80,000; Year 2 2,00,000; Year 3 2,20,000; Year 4 1,50,000. Rate 12%.

FIND. NPV and the decision.

FORMULA. NPV = sum of inflow / $(1 + r)^n$ - cost.

STEPS. $1,80,000 \times 0.8929 = 1,60,714$; $2,00,000 \times 0.7972 = 1,59,439$; $2,20,000 \times 0.7118 = 1,56,592$; $1,50,000 \times 0.6355 = 95,328$. Total = 5,72,072. NPV = total - 5,00,000.

ANSWER. NPV is about Tk. 72,072. It is positive, so the project is acceptable.

CHECK. The plain sum of inflows is 7,50,000, and the discounted total is lower, as expected.

A second case, with messier numbers.

Exceptions and slips. When two projects exclude each other, as when only one oven fits the shop, choose by the higher NPV. When money is limited, rank by profitability. The cost belongs at time zero and is not discounted. Students often use accounting profit instead of cash flow, or forget the cost, or discount at the wrong rate. The discount rate should reflect the risk of the project.

CLOSING THE CHAPTER**SUMMARY**

Capital budgeting judges large, long-lived spending. Its steps run from proposals to review. Only incremental cash flows count. Payback is simple but ignores timing and later cash; NPV counts both and measures added wealth.

THINK IT THROUGH

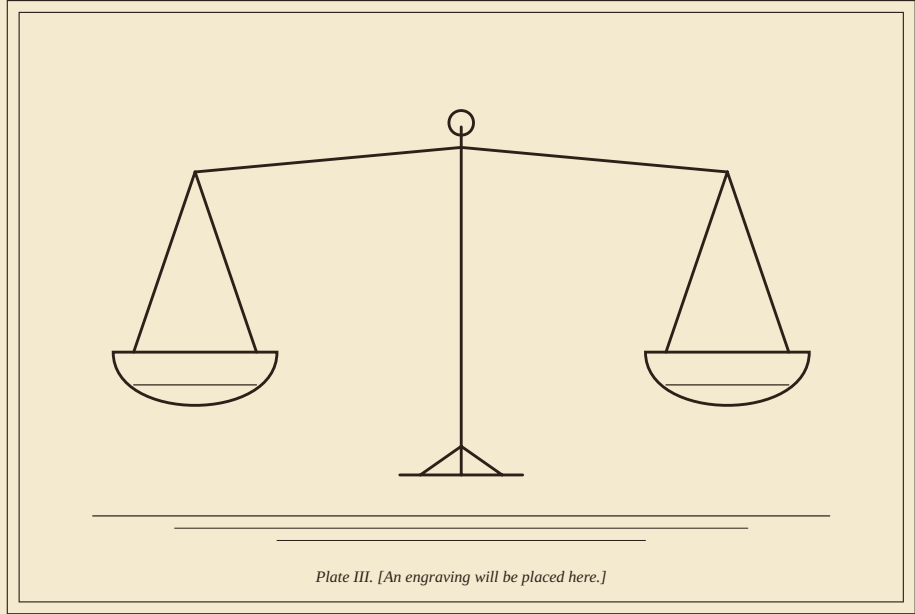
A project costs Tk. 20,000 and returns Tk. 8,000 a year for 3 years. At 10%, find its payback and its NPV. Would you accept it?

WHERE PEOPLE GO WRONG

Students treat sunk costs as relevant and count depreciation as cash. Others prefer the quicker payback although the other project adds more wealth.

IN HISTORY

New York State began the Erie Canal in 1817 and opened it in 1825. It cost about seven million dollars, a vast sum for the state, and was paid for by bor-



CHAPTER III

Judging Projects

A machine costs money now and pays back slowly; is it worth buying?

This chapter explains capital budgeting: why it matters, the steps of the process, which cash flows count, and how to judge projects by payback and by net present value.

The Bakery needs a van but not the cost of one.

A lease is rent for the use of an asset.

Lease. A contract in which the owner of an asset lets another person use it for a fee over a set period.

Lessor. The owner of the asset who lets it to another.

Lessee. The person who uses the asset and pays the fee.

An operating lease is a short rental; a financial lease is a long commitment.

Operating lease. A lease for a short part of the asset's life, which the user can cancel, and where the owner keeps the risks of ownership.

Financial lease. A long lease that covers most of the asset's life and cannot be cancelled, so it works like a loan.

Compare the present value of the after-tax outflows.

A PUZZLE ABOUT A VAN

The Bakery needs a delivery van. Buying costs Tk. 1,00,000 at once. A dealer offers instead to let the Bakery use the van for four years for a yearly fee. Which is better? The answer depends on cost, tax and what remains at the end. To compare fairly, we must put all the cash flows on the same footing.

WHAT A LEASE IS

A **lease** means a contract in which the owner lets another person use an asset for a fee over a set period. The owner is the **lessor**. The user is the **lessee**. Leasing is a form of financing, since the lessee gets an asset without paying its full price now.

TWO KINDS OF LEASE

An **operating lease** lasts for part of the life of the asset, can usually be cancelled, and leaves the lessor with the costs of repairs and with the asset at the end. A **financial lease** covers most of the life of the asset, usually cannot be cancelled, and the lessee bears repairs and similar costs. It works like a loan used to buy the asset.

OPERATING AND FINANCIAL LEASES

Point	Operating lease	Financial lease
Length	Part of asset's life	Most of asset's life
Cancellation	Usually allowed	Usually not allowed
Repairs	Lessor pays	Lessee pays
Total payments	Less than the asset's cost	About the cost plus interest
Best for	Short or changing needs	Long, steady needs

Read each row across: it compares the two leases on one point. The reasons to lease include less money paid at the start, flexibility, and sometimes tax gain. The drawbacks include a higher total cost and, at the end, no ownership.

THE LEASE OR BUY DECISION

The decision has four steps. Step one: find the after-tax cash outflows of leasing. Step two: find those of buying. Step three: discount both at the after-tax cost of debt, because both are in effect ways of borrowing. Step four: choose the plan with the lower present value of outflows. Lease payments reduce taxable profit; so do the interest and depreciation of a purchase. Tax rules differ between countries, so state the rules used.

WORKED EXAMPLE: LEASE OR BUY A VAN

GIVEN. price Tk. 1,00,000; 4 years. Lease payment Tk. 30,000 at each year end. Purchase by a 10% loan with instalments of Tk. 31,500 a year. Depreciation Tk. 25,000 a year. Interest: Year 1 10,000, Year 2 7,500, Year 3 5,000, Year 4 2,500. Tax rate 30% (assumed).

FIND. which plan has the lower present value of after-tax outflows.

FORMULA. After-tax lease cost = payment x (1 - tax rate). After-tax purchase cost = instalment - tax rate x (depreciation + interest). Discount rate = 10% x (1 - 0.30) = 7%.

Before moving on. Multiply by $(1 + r)^n$ to go forward; divide to go back.
Compound interest earns interest on interest.

EQUAL PAYMENTS: ANNUITIES

An annuity is a series of equal payments at equal gaps.

Annuity. A series of equal payments made at equal intervals.

Ordinary annuity. An annuity whose payments are made at the end of each period.

Annuity due. An annuity whose payments are made at the start of each period.

An **annuity** is a series of equal payments made at equal intervals. Rent, loan instalments and regular savings are annuities. In an **ordinary annuity** each payment falls at the end of its period. In an **annuity due** each payment falls at the start. A third kind, the perpetuity, never ends; its present value is the payment divided by the rate.

The formulas are: $FV \text{ of an ordinary annuity} = \text{payment} \times [(1 + r)^n - 1] / r$.
 $PV \text{ of an ordinary annuity} = \text{payment} \times [1 - (1 + r)^{-n}] / r$. For an annuity due, multiply the ordinary result by $(1 + r)$, because every payment comes one period earlier and so earns or loses one more period of interest.

WORKED EXAMPLE: SAVING Tk. 1,000 A YEAR FOR 3 YEARS AT 10%

GIVEN. payment Tk. 1,000 at the end of each year; $r = 10\%$; $n = 3$.

FIND. the future value, the present value, and the future value if payments start at the beginning of each year.

FORMULA. $FV = P \times [(1 + r)^n - 1] / r$; $PV = P \times [1 - (1 + r)^{-n}] / r$.

STEPS. Factor for $FV = (1.331 - 1) / 0.10 = 3.31$, so $FV = \text{Tk. } 3,310$. Factor for $PV = 2.48685$, so $PV = \text{Tk. } 2,486.85$. Annuity due: $3,310 \times 1.10 = \text{Tk. } 3,641$.

ANSWER. The ordinary annuity grows to Tk. 3,310. Its present value is about Tk. 2,487. The annuity due grows to Tk. 3,641.

CHECK. Add each payment by hand: $1,000 \times 1.21 + 1,000 \times 1.10 + 1,000 = 1,210 + 1,100 + 1,000 = \text{Tk. } 3,310$.

PAYMENTS THAT DIFFER: MIXED STREAMS

A mixed stream is handled one payment at a time.

Mixed stream. A series of payments that are not all equal.

A **mixed stream** is a series of payments that are not all equal. No short formula exists. We move each payment separately and add the results. For the present value we divide each payment by $(1 + r)^n$ for its own year. For the future value we multiply each by $(1 + r)$ raised to the years remaining.

WORKED EXAMPLE: PRESENT VALUE OF A MIXED STREAM

GIVEN. Year 1 Tk. 1,000; Year 2 Tk. 2,000; Year 3 Tk. 3,000; discount rate 10%.

FIND. the present value.

FORMULA. $PV = \text{sum of each payment} / (1 + r)^n$.

STEPS. Year 1: $1,000 \times 0.9091 = 909.09$. Year 2: $2,000 \times 0.8264 = 1,652.89$. Year 3: $3,000 \times 0.7513 = 2,253.94$.

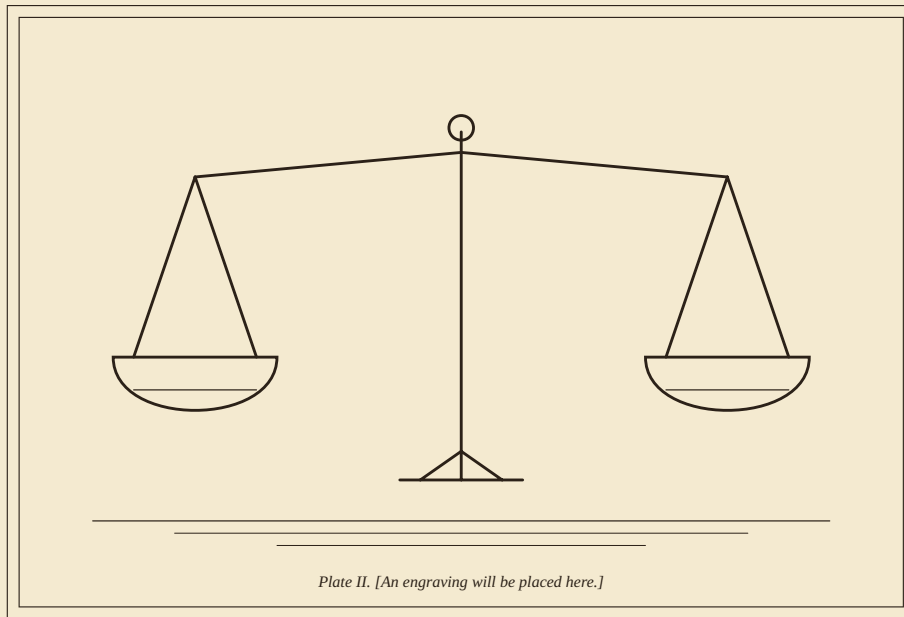
ANSWER. The total present value is Tk. 4,815.93.

CHECK. The total is below the plain sum of 6,000, as it must be, because later money is worth less today.

Exceptions deserve a note. If interest is added more often than once a year, say every month, use r divided by the number of periods and n multiplied by it. When comparing two streams, discount both at the same rate; the stream with the higher present value is worth more today.

Case file, illustrative. The Bakery sets aside Tk. 1,00,000 at the end of each of three years, in an account paying 10%. The ordinary annuity gives $1,00,000 \times 3.31 = \text{Tk. } 3,31,000$, enough toward a second oven.

The case file: saving for a new oven.



CHAPTER II

Money Across Time

A sum in your hand is worth more than the same sum promised later.



This chapter explains why money has a time value, and how to move a sum forward to its future value or back to its present value, for a single sum, for equal payments, and for payments that differ.

² Smith, *The Wealth of Nations*, Book V. The point was formalised by Jensen and Meckling in 1976.

3. $80 \times 2.48685 + 1,000 \times 0.75131 = \text{Tk. } 950.26$; it sells at a discount.

4. A bond is a loan with fixed payments and a repayment date; a share is ownership with no promised payment and no maturity.

CHAPTER VII

1. Repaid within one year; one to about five years; more than five years.

2. $(2/98) \times (360/30) = 24.49\%$.

3. $(3/97) \times (360/35) = 31.81\%$. This exceeds 18%, so take the discount.

4. The delay between sending a payment and the money becoming usable; mail, processing and clearing time.

CHAPTER VIII

1. A new firm needs lasting assets that earn slowly; short loans would fall due before the assets repay them.

2. Term loans, bonds or debentures, preferred stock, common stock, retained earnings.

3. Debt: $15 \times 0.70 = 10.5\%$, cheaper than equity at 18%. Heavy debt raises financial risk, so lenders and owners demand more.

CHAPTER IX

1. Leasing lets one person use an asset owned by another for a fee. Operating leases are short, cancellable, and the lessor bears costs; financial leases are long, not cancellable, and work like loans.

2. After-tax outflows of lease; after-tax outflows of purchase; discount both at the after-tax cost of debt; choose the lower present value.

3. Both plans are in effect borrowing, and interest saves tax, so the after-tax cost of borrowing is the fair rate.

Good finance keeps the business running and growing.

Read each row from left to right: the decision, the question behind it, and a case from a bakery. Later chapters teach how to make each one well.

Finance affects how well a business runs and whether it survives. A firm with the right money at the right time can buy materials, pay workers and keep machines busy. A firm that lacks money may stop production even when customers want its goods. A firm that borrows too much may be unable to pay its lenders. A firm that keeps idle cash earns nothing from it. In each case the quality of financial decisions shows in the firm's efficiency and in its success.

THE GOAL OF THE FIRM

The firm needs one clear aim to judge its decisions.

Profit maximisation. Aiming for the largest possible profit.

Go slowly here.

Decisions need a standard. Without one, we cannot say whether a choice was good. The oldest standard is **profit maximisation**, which means aiming for the largest profit. It sounds sensible, but it has three weaknesses.

Picture two market stalls. Stall A will earn Tk. 100 profit this year and Tk. 100 next year. Stall B will earn Tk. 50 this year and Tk. 200 next year. Which has the larger profit? Stall B, in total. Yet the owner who needs money now may prefer A. The first weakness is that profit maximisation ignores *when* the money arrives. Chapter II explains why money received earlier is worth more.

The second weakness is risk. A profit that is almost certain and a profit that depends on luck count as equal, though they are not. The third is that profit, as accountants measure it, is not the same as cash, as the puzzle showed. A firm can also raise this year's profit by cutting maintenance or training, and so harm the years after.

A better standard is **wealth maximisation**. It means aiming to make the owners' investment as valuable as possible. The value of a firm depends on the cash it will give to its owners in future, the timing of that cash, and the risk of receiving it. This one idea includes the three things that profit leaves out. For this reason, profit is important but is not the ultimate goal.

You should now be able to say: *a firm should choose the plan that adds most to the value of the owners' investment, taking account of time and risk, and not simply the plan with the largest profit this year.*

Pause and check. Why might Stall B be worth more than Stall A, even though the owner who needs cash now prefers A?

OWNERS AND MANAGERS

When owners hire managers, their interests may differ.

Shareholders. The people who own shares, and so own part of a company.

Principal. A person who asks another person to act for them.

Agent. A person who acts on behalf of another.

Agency problem. The risk that agents act for themselves instead of for the owners.

Agency costs. The costs of watching agents and of the losses when

In a large company the owners cannot run the firm. The owners are called **shareholders**. They hire managers to run it for them. In finance the owner is called the **principal** and the manager is called the **agent**. The principal wants the value of the firm to grow. The agent may want a high salary, a comfortable life or a larger firm, even when these do not raise the owners' wealth.

This gap is the **agency problem**. It has a cost. The owners may spend money to watch the managers, for example by hiring auditors, and some bad decisions may still be made. These costs together are called **agency costs**. Owners reduce the problem in two main ways: by linking the manager's pay to the firm's value, for instance through bonuses paid in shares, and by monitoring, through audits, reports and a board of directors.

Exceptions matter. In a business run by its single owner there is no agency problem, since principal and agent are the same person. The problem also exists

Face value. The sum printed on the bond and repaid at maturity; also called par value. (Chapter VI)

Finance. The way a business finds, uses and controls its money. (Chapter I)

Financial lease. A long lease that covers most of the asset's life and cannot be cancelled, so it works like a loan. (Chapter IX)

Financial leverage. The effect of fixed financing costs, such as interest, on how profit after interest responds to EBIT. (Chapter V)

Financial manager. The person who decides how a firm finds and uses its money. (Chapter I)

Financial risk. The extra risk that comes from using borrowed money. (Chapter IV)

Financing decision. A choice about how to raise money, from owners or from lenders. (Chapter I)

Fixed cost. A cost that stays the same whatever the level of sales. (Chapter V)

Float. The time between a payment being sent and the money becoming usable by the receiver. (Chapter VII)

Future value. What a sum of money today will be worth at a later date, after earning interest. (Chapter II)

Incremental. Extra; the change in cash caused by choosing the project. (Chapter III)

Initial public offering. The first sale of a company's shares to the public. (Chapter VIII)

Interest. The payment a borrower makes for the use of money, shown as a percentage of the sum. (Chapter II)

Intrinsic value. What a security is worth, found from the present value of the cash it is expected to pay. (Chapter VI)

Investment decision. A choice about which assets, such as machines or shops, the firm should buy. (Chapter I)

Lease. A contract in which the owner of an asset lets another person use it for a fee over a set period. (Chapter IX)

Lessee. The person who uses the asset and pays the fee. (Chapter IX)

Lessor. The owner of the asset who lets it to another. (Chapter IX)

Leverage. The use of fixed costs to magnify the effect of a change in sales on profit. (Chapter V)

Liquidity. How quickly and easily an asset can be turned into cash without loss. (Chapter VII)

Matching principle. Financing an asset for about as long as the asset will be used. (Chapter VII)

Maturity. The date on which the bond is repaid. (Chapter II)

Mixed stream. A series of payments that are not all equal. (Chapter II)

Net present value. The present value of a project's cash inflows minus its initial cost. (Chapter III)

Operating lease. A lease for a short part of the asset's life, which the user can cancel, and where the owner keeps the risks of ownership. (Chapter IX)

Operating leverage. The effect of fixed operating costs on how profit before interest and tax changes with sales. (Chapter V)

Ordinary annuity. An annuity whose payments are made at the end of each period. (Chapter II)

Payback period. The time a project takes to return its initial cost in cash. (Chapter III)

Preferred stock. Shares that pay a fixed dividend and rank before ordinary shares, usually without a vote. (Chapter VIII)

Present value. What a sum due in the future is worth today. (Chapter II)

Principal. A person who asks another person to act for them. (Chapter I)

Probability distribution. A list of possible returns, each with the chance that it occurs. (Chapter IV)

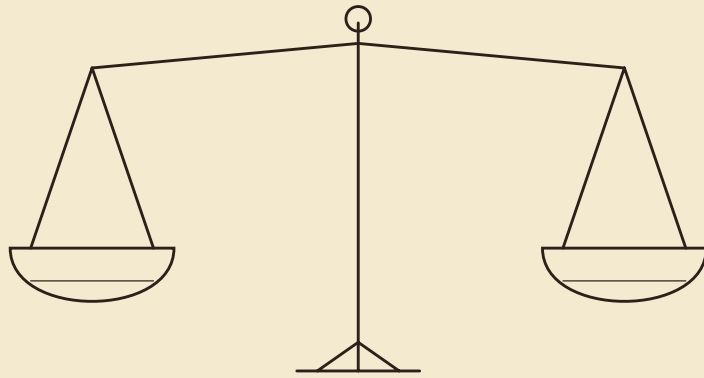


Plate I. [An engraving will be placed here.]

CHAPTER I

The Purpose of Finance

Money is easy to count and hard to use well.



This chapter explains what finance is, what a financial manager decides, why the aim of a firm is wealth and not only profit, and why owners and managers may want different things.

CONTENTS

I	THE PURPOSE OF FINANCE	1
II	MONEY ACROSS TIME	1
III	JUDGING PROJECTS	1
IV	RISK AND RETURN	1
V	LEVERAGE	1
VI	VALUING BONDS AND SHARES	1
VII	SHORT-TERM MONEY	1
VIII	LONG-TERM FUNDS	1
IX	LEASING	1
	ANSWERS TO REVIEW QUESTIONS	1
	GLOSSARY	1
	FURTHER READING	1
	MATHS YOU WILL USE	1



PRINCIPLES TO FINANCE



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An Introduction for First-Year Students*

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